



What does this get us?

4

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The operational semantics, and its implementation, are sound with respect to the axiomatic model

$$\begin{array}{c}
\frac{}{\text{when}(\bar{c})\{s_1\}; s_2 \hookrightarrow s_2 \mid (\bar{c}, s_1)} \text{STEP} \\
\\
\frac{s_1 \hookrightarrow s_2 \mid (\bar{c}_3, s_3) \quad j \text{ fresh}}{\langle \bar{b}_r[(i, \bar{c}_1, s_1)], \bar{b}_p \rangle / H \rightsquigarrow \langle \bar{b}_r[(i, \bar{c}_1, s_2)], \bar{b}_p : (j, \bar{c}_3, s_3) \rangle / H[i += S_j]} \text{SPAWN} \\
\\
\frac{C = \{c \mid c \in \bar{c}' \wedge (_, \bar{c}', _) \in \bar{b}_r \cup \bar{b}_{p1}\} \quad C \cap \bar{c} = \emptyset}{\langle \bar{b}_r, \bar{b}_{p1} : (i, \bar{c}, s) : \bar{b}_{p2} \rangle / H \rightsquigarrow \langle \bar{b}_r : (i, \bar{c}, s), \bar{b}_{p1} : \bar{b}_{p2} \rangle / H[i += R_i][\bar{c} += R_i]} \text{RUN} \\
\\
\frac{}{\langle \bar{b}_r[(i, \bar{c}, \text{done})], \bar{b}_p \rangle / H \rightsquigarrow \langle \bar{b}_r[\epsilon], \bar{b}_p \rangle / H[i += C_i][\bar{c} += C_i]} \text{COMPLETE}
\end{array}$$

■ **Figure 9** Semantics of the core BoC calculus with histories.

$$\begin{array}{c}
\frac{}{i \vdash_b \epsilon} \text{WFB-EMPTY} \quad \frac{}{i \vdash_b R_i : \bar{S} : (C_i?)} \text{WFB-NEMPTY} \\
\\
\frac{}{\vdash_c \epsilon} \text{WFC-EMPTY} \quad \frac{}{\vdash_c R_i} \text{WFC-SINGLE} \quad \frac{\vdash_c \bar{E}}{\vdash_c R_i : C_i : \bar{e}} \text{WFC-PAIR}
\end{array}$$

■ **Figure 10** Well-formedness rules for behaviour and cown histories

4.3 Soundness of Core Calculus According to the Axiomatic Model

In the previous section, we showed that a well-formed history translates to a valid execution in the axiomatic model. In this section we show that the semantics is sound according the axiomatic model by proving that it always produces a well-formed history.

In addition to general well-formedness, the history of an execution is going to match the specific configuration that it was produced together with.

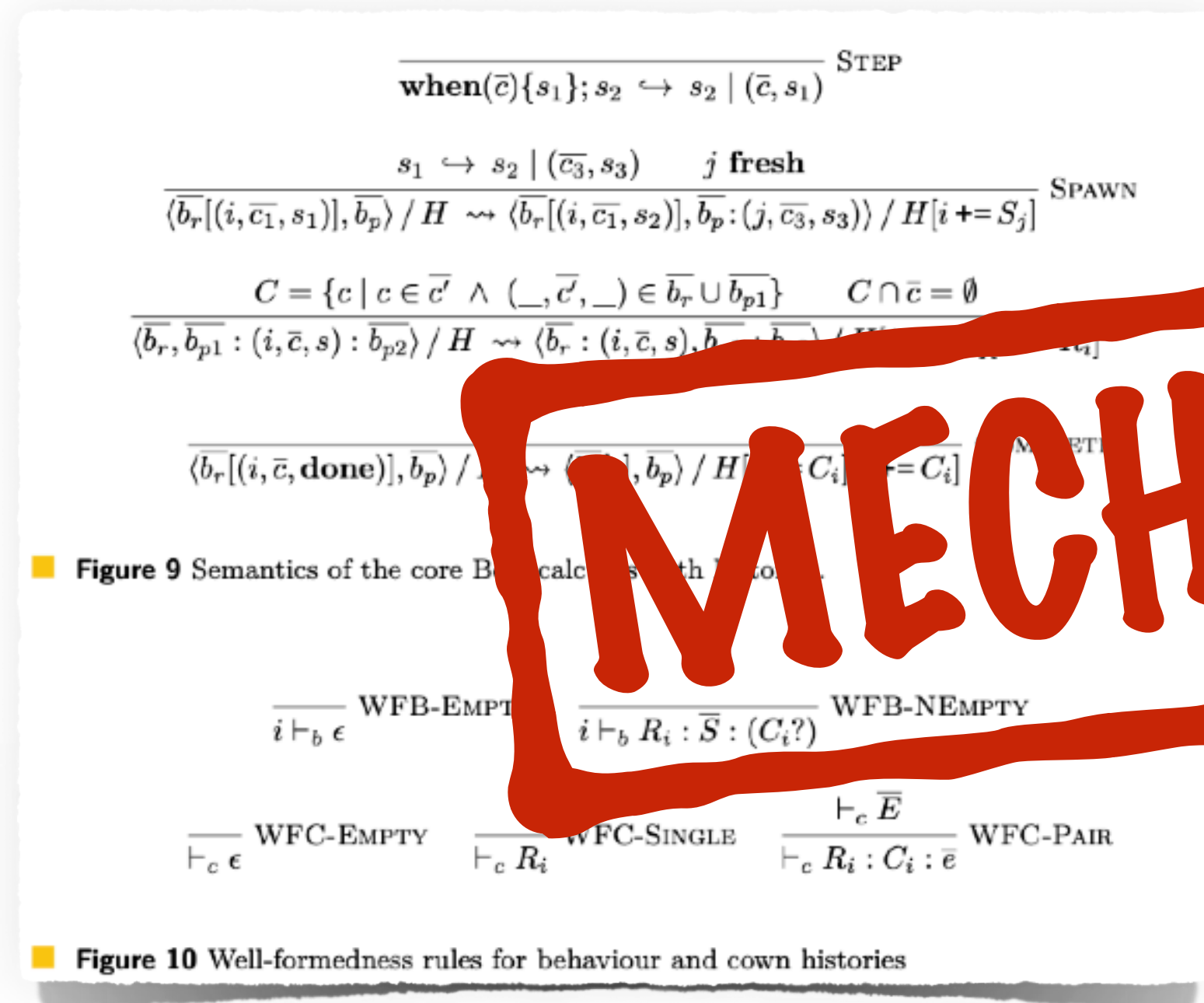
► **Theorem 15** (Preservation of well-formed and matching histories). *Starting from a well-formed and matching history, evaluation produces another well-formed and matching history:*

$$\tau \vdash H \wedge H; \tau \vdash c/g \wedge c/g/H \rightarrow c/g/H' \implies \exists \tau'. \tau' \vdash H' \wedge H'; \tau' \vdash c/g'$$

Proof. By cases on the evaluation relation. For each case we select a τ' that is τ extended so that the new event produced by that evaluation step is given a timestamp larger than any previous timestamp. Preservation of each of the two relations can then be proved separately. We refer to the mechanisation for details [7]. ◀

MECHANISED

What does this get us?



4.3 Soundness of Core Calculus According to the Axiomatic Model

In the previous section, we showed that the operational semantics translates to a valid execution in the axiomatic model. In this section we show that the axiomatic semantics is sound according to the operational model by proving that it also translates to a well-formed history.

In addition to this, we show that, the state of the execution is going to match the specific configuration that it is associated together with.

MECHANISED

Theorem 15 (Preservation of well-formed and matching histories). *Starting from a well-formed and matching history, evaluation produces a well-formed and matching history:*

$$i \vdash_b \epsilon \quad H = \langle H, H \rangle \rightarrow \langle H', H' \rangle \Rightarrow \exists \tau'. \tau' \vdash H' \wedge H'; \tau' \vdash \langle H', H' \rangle$$

Proof. By cases on the evaluation relation. For each case we select a τ' that is τ extended so that the new event produced by that evaluation step is given a timestamp larger than any previous timestamp. Preservation of each of the two relations can then be proved separately. We refer to the mechanisation for details [7].

The operational semantics, and its implementations, are sound with respect to the axiomatic model

What does this get us?

Update path (src then dst)

```
when(src) { A src.balance += 10; }  
when(dst) { B dst.balance += 10; }
```

Diagnostic path (dst then src)

```
when(dst) { C print(dst.balance) }  
when(src) { D print(src.balance) }
```

	src=0, dst=0	src=0, dst=10	src=10, dst=0	src=10, dst=10
Order	C C C D D D	B B B D D D	A A A C C C	A A A B B B
	B D D A A A	C D D A A B	C C D C A A	B B D C A C
	D C B A C A	C A C A B C	B D C B D B	C D B C D C
	A B A C C C	A A B C C A	D B B D B A	D C C D C A
	A B A C C C	A A B C C A	D B B D B A	D C C D C A
	A B A C C C	A A B C C A	D B B D B A	D C C D C A
Permitted	✓	⚠	✓	✓